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[ ]: prn - 12412485
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[2]: import pandas as pd
import matplotlib.pyplot as plt
```

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[9]: df = pd.read_csv("Marks.csv")
df
```

```
[9]:
```

	Roll	Sub 1	Sub 2	Sub 3	Sub 4	Sub 5
0	1	80	87	97	92	74
1	2	72	89	81	86	85
2	3	99	83	99	77	94
3	4	70	92	76	79	70
4	5	92	79	84	85	89
...	...	...	...	...	...	...
70	71	71	80	93	74	78
71	72	86	74	82	94	84
72	73	73	79	94	72	77
73	74	75	91	79	91	79
74	75	91	88	93	78	89

75 rows x 6 columns

```
[12]: def median(arr):
    n = len(arr)
    # med = 0
    if (n%2==0):
        return (arr[n//2] + arr[n//2 - 1])/2
    else:
        return arr[n//2]
    # return med

def mean(arr):
    n = len(arr)
    ans = 0
    for num in arr:
        ans += num
    return ans/n

def mode(arr):
    a = np.array(arr)
    return a.mode()

def variance(arr):
    n = len(arr)
    ans = 0
    x = mean(arr)
    for num in arr:
        ans += (num-x)**2
    return ans/n

def standardDeviation(arr):
    return variance(arr)**0.5
```

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[10]: df = df.iloc[:, 1:]
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```
stats = []
labels = []

for col in df.columns:
    data = df[col].dropna()

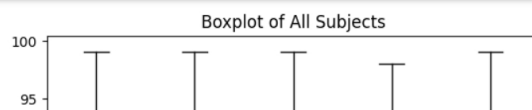
    q0 = data.min()
    q1 = data.quantile(0.25)
    q2 = data.quantile(0.5)
    q3 = data.quantile(0.75)
    q4 = data.max()

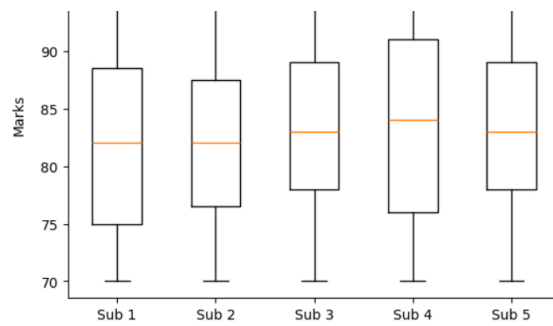
    stats.append({
        "med": q2,
        "q1": q1,
        "q3": q3,
        "whislo": q0,
        "whishi": q4,
        "fliers": []
    })

    labels.append(col)

fig, ax = plt.subplots()
ax.bxp(stats, showfliers=False)
ax.set_xticklabels(labels)
ax.set_title("Boxplot of All Subjects")
ax.set_ylabel("Marks")

plt.show()
```

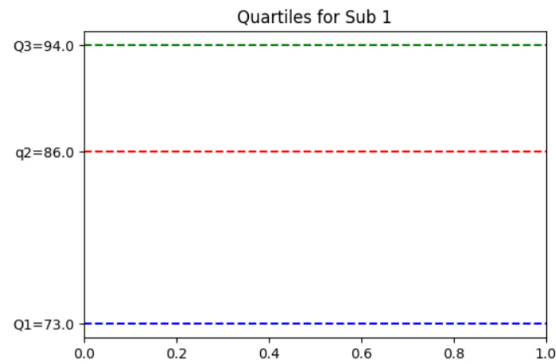




```
[19]: # q1 = df['Sub 1'].quantile(0.25)
# q2 = df['Sub 1'].quantile(0.5)
# q3 = df['Sub 1'].quantile(0.75)
# print(q1)
# print(q2)
# print(q3)

# plt.figure(figsize=(6,4))
# plt.hlines([q1, q2, q3], xmin=0, xmax=1, colors=['blue','red','green'], linestyle='--')
# plt.xlim(0,1)
# plt.title("Quartiles for Sub 1")
# plt.yticks([q1, q2, q3], [f"Q1={q1:.1f}", f"Q2={q2:.1f}", f"Q3={q3:.1f}"])
# plt.show()
```

```
73.0
86.0
94.0
```



```
[11]: quartiles = df[['Sub 1','Sub 2','Sub 3','Sub 4','Sub 5']].quantile([0, 0.25, 0.5, 0.75])
print(quartiles)
```

	Sub 1	Sub 2	Sub 3	Sub 4	Sub 5
0.00	70.0	70.0	70.0	70.0	70.0
0.25	75.0	76.5	78.0	76.0	78.0
0.50	82.0	82.0	83.0	84.0	83.0
0.75	88.5	87.5	89.0	91.0	89.0

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